Leveraged Buyouts & Value Creation (Session 9)

> **Exam scope updated 14 October 2026.** The professor excludes money market hedge (p 88), currency swaps (pp 109 to 117), DCF valuation and guest speaker lectures from the half course exam and the Part I questions of the full exam. Everything in this fiche is inside the scope. Source: Blackboard announcement "Exam content clarifications", 13 October 2026. Optional reading: HBS note "Note on LBO Capital Structure" (Ivashina).

One session, one deal. Learn what an LBO is, where the return comes from, how the model is built in eight steps, and how to tell a safe structure from a fragile one.

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1. Private equity in one page

Private equity is equity invested in unlisted companies, held for years, sold for a gain. It is illiquid, it usually comes with control, and only the exit value counts.

- **LP (limited partners)** commit the capital: pension funds, endowments, insurers, family offices. Funds live 10-12 years.
- **GP (general partner)** picks and manages the deals, for **1-2% management fees + 20% carried interest** above a hurdle (the carry is earned only once LPs got their minimum return).
- Where LBOs sit in the life cycle: the **maturity phase**, when cash flows are stable enough to carry debt. Start-ups run on equity; turnarounds are a different fund.

> **Remember:** an LBO fund buys mature, predictable businesses because their cash flow is what repays the acquisition debt.

2. The three engines of value creation (A-L-O)

Every euro of sponsor return comes from one of three engines. On an exam, name the engine before you calculate.

Engine	What it is	One-line example
A - Arbitrage	Buy low, sell high: the exit multiple beats the entry multiple	In at 7x, out at 9x with the same EBITDA
L - Leverage	The target's cash flow repays the debt, turning enterprise value into equity value	Debt falls 300 to 150 while EV is flat
O - Operational	Margins and revenue improve: the business itself is worth more	EBITDA grows 80 to 100 at a constant multiple

> **Remember:** only A and O create value. Leverage creates nothing by itself: it amplifies whatever A and O produce, gains and losses alike.

3. What an LBO is, and what makes a good target

An **LBO** is an acquisition financed mostly with debt: the sponsor writes a thin equity cheque (typically 20-40%), lenders provide the rest (60-80%), and **the target's own cash flow repays the debt** over a ~5-year hold. The sponsor then sells at a higher EBITDA, a higher multiple, or simply with less debt.

A good target, in three groups:

- **Cheap to buy:** undervalued vs peers, under-leveraged, excess cash on the balance sheet.
- **Room to improve:** margins or growth lag peers, a catalyst is waiting, management can execute (or be upgraded).
- **Can carry debt:** stable and predictable cash flows, high barriers to entry, moderate capex and R&D, assets that could be sold if needed.

> **Remember:** a biotech startup with negative, volatile cash flow is the worst LBO candidate: no cash flow, no debt capacity, no deal.

4. Why leverage amplifies the equity return

Same company: bought 100, sold 180 five years later, debt held constant. Only the funding mix changes.

	Equity in	Equity out	MOIC
All-equity	100	180	1.8x
50% debt	50	130	2.6x
70% debt	30	110	3.7x

Two mechanisms stack up:

- Each euro of debt repaid converts enterprise value into equity value, automatically.
- A smaller equity cheque means the same value creation lands on fewer euros of equity.

The mirror image: if the exit value falls, leverage multiplies the loss just as fast.

5. The players and the four exits

- **Sponsor** (the PE fund): equity and control. **Lenders**: banks and bond investors, the debt tranches. **NewCo/BidCo**: the vehicle that borrows, buys, then merges with the target. **Management**: often rolls over equity and runs the plan.

Exit	What happens	Note
Strategic sale	Sold to a corporate buyer	Often the best price: synergies
Secondary buyout	Sold to another sponsor	The next fund sees more value to create
IPO	Listed, shares sold down over time	Needs a receptive market
Dividend recap	The company re-levers and pays a dividend	Cash back without a full exit

6. The LBO model in eight steps

One disciplined sequence. Each step feeds the next.

Step 1 - Price the deal. Private target: Entry EV = entry multiple x normalised EBITDA. Our deal: EBITDA 409.0 x 8.5 = EV 3,476.5 (no existing net debt).

Step 2 - Sources & uses. Uses: purchase price 3,476.5 + fees 52.1 (1.5%) = 3,528.6. Sources: TLB 1,308.8 + senior notes 613.5 + subordinated notes 409.0 = debt 2,331.3 (66%) + **sponsor equity 1,197.3 (34%)**. Equity is always the plug: it fills whatever debt does not fund, and the two columns must balance.

Step 3 - Set the debt stack. Cheapest and most senior at the top, sized in turns of EBITDA:

Layer	Pricing	Profile
Revolver (RCF)	SOFR + 200	Senior, undrawn at close, working-capital backstop
Term Loan A	SOFR + 300	Senior secured, amortises ~20%/yr, ~5 yrs
Term Loan B	SOFR + 350	Senior secured, ~1%/yr + bullet, ~7 yrs
Senior notes	~10% fixed	Unsecured, bullet, callable at a premium
Subordinated notes	~12% fixed	Longest, first loss after equity
Sponsor equity	the residual	Junior, first loss, all the upside

Pricing steps up about 100 bps per junior layer, plus ~50 bps for longer maturity. Total leverage here: 5.7x EBITDA.

Steps 4-5 - Project and derive the cash. Project the P&L (growth, margins), then:

> Levered FCF = EBITDA - maintenance capex - change in NWC - cash interest - cash taxes

Step 6 - The debt schedule and the cash sweep. Cash goes out in strict order:

1. Mandatory amortisation first: TLA ~20%/yr, TLB ~1%/yr.
2. The sweep prepays the TLA until it is gone.
3. Then the sweep prepays the TLB.
4. Bonds stay outstanding (bullets, callable at a premium); leftover cash builds up.

Our deal: debt falls 2,331.3 to 1,668.8 in five years (5.7x to 3.7x EBITDA) - only the TLB moves, from 1,308.8 to 646.3.

Step 7 - Circularity. Interest depends on debt, debt depends on the sweep, the sweep depends on interest. Enable iterative calculation and let the loop settle.

Step 8 - Returns. Next two sections.

7. Multiples: choose them and apply them

Multiple = price of the asset / profit generated by the asset

Multiple	Use it when	Watch for
EV/EBITDA	The LBO default; cyclical sectors	Ignores capex and tax differences
EV/EBIT	Capital-heavy, comparable depreciation	Accounting policies
P/E	Profitable companies, forward EPS available	Distorted by leverage
EV/Sales	Negative earnings, early stage	Partial picture only
Price/Book	Financials, capital-intensive	Accounting-driven

Three working rules:

- **Consistency first.** EV pairs with asset-side profits (EBITDA, EBIT); equity price pairs with equity-side profits (net income, EPS). Mixing the two is the classic error.
- **Define before you compare.** Trailing vs forward, basic vs diluted, before vs after extraordinary items.
- **Peers on fundamentals, not labels.** Same growth and margins, or not a peer.

> **Example (Bulgari-LVMH):** screening 15 luxury names on category, growth and margins keeps only the true peers. Relevant comps value Bulgari at 7.3-13.7 per share against 5.6-9.7 on the full sample. The peer set moves the price more than the method.

8. Returns: IRR, MOIC and the five levers

- **IRR** is annualised: it sees timing. **MOIC** (cash-on-cash) is cash in divided by cash out: it ignores timing.
- Report both. The same 1.8x reached one year earlier means a higher IRR.

Base case of our deal: **12.1% IRR / 1.8x MOIC** (8.5x in, 8.5x out, five years). Exit a year early: 9.6%. Exit one turn lower (7.5x): 6.9%.

The five levers, mapped to the engines:

Lever	Engine	Direction
Purchase price	A	Pay less, earn more: the strongest lever
Exit multiple	A	Sell higher
Leverage	L	Smaller cheque, same outcome split harder
Revenue growth	O	Bigger EBITDA at exit
EBITDA margin	O	More cash per euro of sales

9. A full deal, entry to exit (class workbook)

One clean pass through the model, numbers rounded. Entry EBITDA 100, entry multiple 10x, existing net debt 30, fees 2% of EV.

- **Step 1 - Price it.** EV = 100 x 10 = 1,000. Seller proceeds = 1,000 - 30 = 970.
- **Step 2 - Fund it.** Uses = 970 + 30 (refinance) + 20 (fees) = 1,020. New debt = 4 x 100 = 400. Sponsor equity = 1,020 - 400 = **620** (the plug).
- **Step 3 - Project.** Revenue 250 growing 10%/yr at a 40% EBITDA margin. Exit EBITDA = 250 x 1.1^5 x 40% = **161.1**.
- **Step 4 - Repay.** Taxes 40%, capex 4.5% of revenue, NWC 15% of the revenue increase, interest 10% on beginning debt. Debt falls 400 to **173.3** by exit.
- **Step 5 - Exit at 11x.** EV = 161.1 x 11 = 1,771.6. Sponsor proceeds = 1,771.6 - 173.3 = **1,598.2**.
- **Step 6 - Measure.** MOIC = 1,598.2 / 620 = **2.58x**. IRR = 2.58^(1/5) - 1 = **20.9%**.

Where does the 978.2 of gain come from? The attribution must reconcile (gains shown to the cent):

Driver	Calculation	Gain
EBITDA growth	(161.05 - 100) x 10	610.51
Multiple expansion	(11 - 10) x 161.05	161.05
Debt repaid	400 - 173.32	226.68
Fees	entry fee	-20.00
Total		**978.24**

Sensitivity is not a promise: at 4x leverage, IRR runs 15.5% to 23.2% for exit multiples 9x to 12x; at the 11x exit, 3x leverage gives 19.2% and 7x gives 30.2%, with proportionally more downside risk.

10. Is the structure safe?

Three ratio families, tested every year. A healthy LBO gets safer as it ages.

- **Leverage:** Net debt/EBITDA (safe: at or below ~6x) and Senior secured/EBITDA (safe: at or below ~4x).
- **Coverage:** EBITDA/interest and (EBITDA - capex)/interest.
- **Liquidity:** cash on hand + undrawn revolver.

Our deal: net debt/EBITDA falls 5.6x to 3.7x over the hold - inside the safe zone and improving.

11. Exam traps

- Saying leverage creates value. It amplifies; only A and O create.
- Forgetting equity is the plug in sources & uses.
- Sweeping the bonds: the sweep prepays term loans only; notes are bullets, callable at a premium.
- Mixing numerator and denominator (EV/net income, equity value/EBITDA).
- Comparing multiples with different definitions (trailing vs forward, basic vs diluted).
- Building peers on sector labels instead of growth and margins.

• Targeting volatile, capex-heavy, early-stage companies for an LBO.
• Reading MOIC as if it captured timing; that is what IRR is for.

12. Must know absolutely

Must know:
• LP commits, GP manages for 1-2% + 20% carry over a hurdle; funds live 10-12 years; LBOs live in the maturity phase.
• A-L-O: arbitrage (multiple), leverage (paydown), operational (margins). Only A and O create value.
• LBO = 60-80% debt, the target's cash flow repays it, exit in ~5 years.
• Good target: cheap, improvable, able to carry debt (stable cash flows, moderate capex).
• Eight steps: price, sources & uses (equity is the plug), tranches, P&L, levered FCF, debt schedule with sweep (TLA then TLB, bonds are bullets), circularity, returns.
• IRR sees timing, MOIC sees scale; report both.
• Five levers: price, exit multiple, leverage, growth, margin.
• Safety: leverage at or below ~6x, coverage, liquidity; a good deal de-levers with age.